3 FIRE DETECTION

Property 3 of 7

THE PROBLEM

Most fires start small. The danger is that nobody sees them in time. Early detection gives people time to leave safely.

YOUR MISSION

Find the early signs of a fire, sound a clear alarm and help reduce the danger.

TARGETS AND LIMITS

Target	Requirement
Levels	The system shows three danger levels, for example normal, warning and alarm, on the dashboard and on the board.
Response	The alarm starts within 20 seconds after the test condition is applied.
Reliability	No alarm in normal conditions during the whole demo.
Reset	Only an allowed person can reset the alarm.

IN THE REAL WORLD

A real building has many rooms and strict safety rules. A real system is certified, tested often and must keep working when a part fails.

ADVANCED OBJECTIVES (OPTIONAL)

- Estimate the risk level and how fast it rises
- Detect a faulty sensor
- Watch two rooms on one dashboard

SAFETY RULES

No real fire, flame or smoke is allowed at the event. The organizers will show you safe ways to test.

TEAM PLANNING SHEET

Use this sheet during Build Your Strategy. You have 45 minutes. Write short and clear answers. Submit it with the first link sent to your team's email. The jury checks it at the plan check.

Each team receives two links by email. The first link is for your plan, during Build Your Strategy. The second link is for your presentation, at the end of Final Inspection, before you present to the jury.

Team name: Cappucin	**Property:** 3 - Fire Detection (project: EMBER)
Members: Israa Soudani, Mayssam Hergli, Moetez Zouari	**Team leader:** Moetez Zouari
Constraint card:	

THE PROBLEM IN OUR OWN WORDS

Who is affected? What can fail? What does an incident or a loss cost?

Who is affected: families in homes that cook with butane bottles, small restaurants and workshops, and the building managers responsible for them.
What can fail: a small fire or gas leak starts at night or in an empty room and nobody sees it in time. Classic detectors give false alarms, so people ignore or unplug them.
What it costs: lives and injuries first, then the home or business, its stock and income. Every minute of early warning gives people more time to leave.

OUR SIX LAYERS

Layer	What we will do	Who is in charge
Sensing	Three sensors of different kinds: gas sensor (smoke and combustible gas), DHT22 (temperature, used for how fast heat rises), LDR sampled fast (flame flicker). Plausibility checks flag a faulty or disconnected sensor: DHT22 returns no value, gas reading stuck at 0 or max, LDR flat. One odd reading never starts an alarm: decisions use a 5 s window.	
Communication	The ESP32 sends a reading packet every 0.5 s over Wi-Fi to the laptop gateway, and receives commands back. Traffic is protected with HTTPS (TLS, pinned certificate). Every command is also signed with HMAC-SHA256 plus a counter, so a copied or replayed command is rejected.	
Edge processing and AI	A logistic regression model runs on the ESP32. It reads a 5 s window (10 readings) and computes 6 features: gas above baseline, gas slope, heat slope, light change, flicker, camera flame score (optional, laptop OpenCV). Output: probability of normal, gas leak, fire, mapped to NORMAL / WARNING / ALARM. Trained on data we record, including traps (flashlight, warm hand). Tested on at least 20 labeled windows with a confusion matrix, accuracy, precision and recall. Key metric: fire recall. Compared to a simple threshold rule.	
Actuation	Automatic: LEDs green, yellow, red; buzzer with one pattern per level; servo as the gas shut-off valve, closed at gas leak or fire. Manual: an allowed user can open or close the valve and silence the buzzer from the dashboard. Safe state on failure or lost link: valve closed, local alarm keeps running.	
Security	Threat 1: someone resets or silences the alarm without the right to (prank, intruder). Protection: dashboard login, reset needs an authorized account, the board's button can only silence for 60 s, never reset. Threat 2: fake or replayed commands on the network (for example "open valve"). Protection: HMAC-SHA256 signature + counter, rejected and logged. Threat 3: eavesdropping or tampered data in transit. Protection: TLS. Threat 4: connection lost or jammed. Protection: the board decides and acts alone.	
Operator interface	Web dashboard on the laptop: live values, a big colored status banner (NORMAL / WARNING / ALARM), fire probability with "why" bars, sensor fault indicators, event history, commands (authorized reset, silence, valve open or close). A non-expert reads it in a few seconds.	

OUR ARCHITECTURE

Sketch the six layers and show where each target of our property is handled (architecture diagram).

T1 Levels: model output on the ESP32, shown on the LEDs and the dashboard banner.
T2 Alarm within 20 s: 0.5 s loop on a 5 s window. Our target: under 10 s.
T3 No alarm in normal conditions: model trained on traps (flashlight, warm hand) + 5 s window against odd readings.
T4 Only an allowed person resets: dashboard login + signed reset command; the board button only silences for 60 s.
Architecture diagram: bottom of the next page.

TEAM PLANNING SHEET (CONTINUED)

OUR PARTS LIST

Part	How many	Cost (credits)	Why we need it	Alternative if sold out
ESP32 development board	1	150	Brain, Wi-Fi, runs the AI model	Arduino Uno (100), model runs on the laptop over USB
Gas sensor	1	90	Smoke and combustible gas	None in the bank: DHT22 + LDR still give two sensors
DHT22 sensor	1	70	Temperature, how fast heat rises	Gas + LDR still give two sensors
LDR light sensor	1	30	Flame flicker	Infrared obstacle sensor (40)
Servo motor	1	70	Gas shut-off valve, closes by itself	Small fan (80) as a vent, gas leak only
Buzzer	1	20	Siren, one pattern per level	Dashboard siren + red LED strobe on the board
LED, red + yellow + green	3	30	Shows the danger level on the board	Fewer LEDs + level shown in the buzzer pattern
Push button + extra wires, male-female	1 + 1 pack	10 + 5	Local silence for 60 s, never a reset; spare wiring	Dashboard silence only; starter pack wires only

Total cost: 475 credits **Our budget:** 490 credits (440 after a re-draw) **Estimated unit cost of the real deployment:** **about 160-260 DT per room** (detector + gas valve; rough estimate; gateway, certification and installation extra)

HOW OUR PROTOTYPE MAPS TO THE REAL SOLUTION

What changes, what is preserved, and why the prototype stays representative.

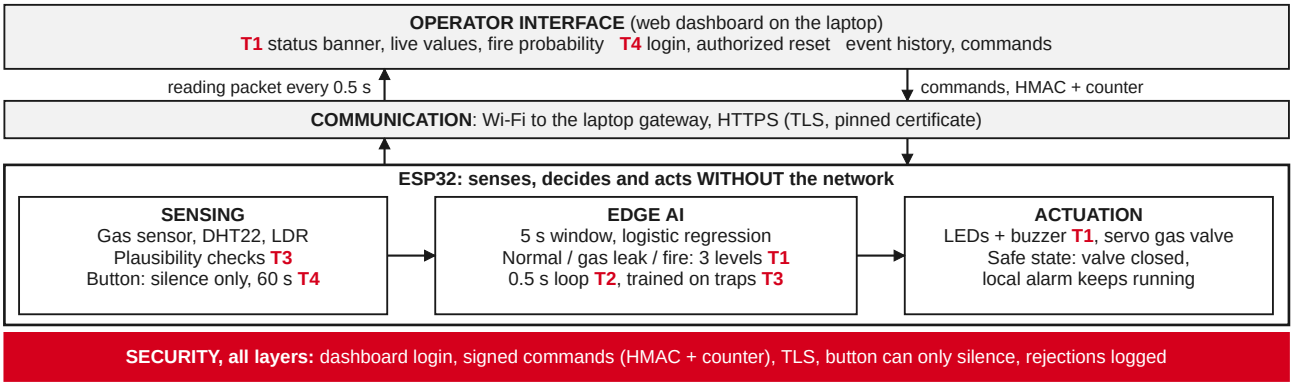
What changes: LDR → infrared flame sensor; hobby gas sensor → certified gas and smoke sensor; servo → gas solenoid valve that closes without power; laptop → building gateway + cloud dashboard; local Wi-Fi → building Wi-Fi or LoRa with TLS; one USB-powered room → one detector per room on mains + battery; safe test condition → real smoke, heat and flame.
Added in the real product: voice alerts that say where the danger is; interconnection, so all detectors in the building alarm together; building actions (gas closed, ventilation off, room power cut); automatic call to 198 with a spoken brief (room, floor, intensity, gas hazard).
What is preserved: the three early signs; the same 5 s model on the detector, three levels, leak vs fire; acts without internet, safe state; login, signed commands, encrypted traffic, authorized reset.
Why it stays representative: same decision logic, data flow and security; better sensors and valve, model retrained on real data.

OUR BIGGEST RISK AND OUR PLAN B (ALSO FOR SOLD-OUT PARTS)

Biggest risk: the model meets conditions it was not trained on, giving a false or missed alarm in the demo: the organizers set the safe test condition and the gas sensor drifts until warm.
Plan B: ask for the test condition first and record training data on exactly that, plus traps (flashlight, warm hand); power the gas sensor 20+ minutes before recording or demo; the 5 s window rejects odd readings; rehearsed backup demo mode if hardware fails.
Network failure: the board decides and acts alone; phone hotspot for the link to the laptop if the venue Wi-Fi fails.
Sold-out parts: we ask first for the gas sensor (4 in stock) and the buzzer (3). Every part has an alternative above, within 490 credits.

OUR ARCHITECTURE (DIAGRAM)

T1 Levels **T2** Response **T3** Reliability **T4** Reset



SECURITY, all layers: dashboard login, signed commands (HMAC + counter), TLS, button can only silence, rejections logged